# The Double Manifold + an application

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# Context: spectral inequalities with boundary

Let $(M, g)$ be a compact Riemannian manifold **with boundary**, with Dirichlet or Neumann boundary conditions. Denote by $(e_k)$ the eigenfunctions of $-\Delta$ with eigenvalues $\lambda_k^2 \to +\infty$.

**Goal**: extend spectral inequalities to the case with boundary.

Previous results (Logunov–Malinnikova, Jerison–Lebeau) apply to *boundaryless* manifolds. To handle the case with boundary:

## Central idea

Reduce the problem with boundary to the boundaryless case via the **double manifold** $\widetilde{M}$.

# The three challenges

The construction must be carried out carefully. The main obstacles are:

1. Low regularity: the metric is $W^{1,\infty}$ and the domain is $W^{2,\infty}$.
2. No geodesic coordinates: the classical reflection requires $C^2$ (resp. $C^3$) regularity for the metric (resp. the domain). At low regularity, the geodesic flow may not be $C^1$.
3. The glued metric must be Lipschitz: gluing two copies must produce a $W^{1,\infty}$ metric on the double manifold.

**Solution**: construct a *pseudo-geodesic system* of class $W^{2,\infty}$ that is tangent to the geodesic coordinate system at the boundary.

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# Definition of the double manifold

## Definition

*The* **double manifold** *of $M$ is the topological space*

$$\widetilde{M} = \overline{M} \times \{-1, 1\} \,/\, \partial M,$$

*where we identify the boundary points $(x, -1) \sim (x, 1)$ for all $x \in \partial M$.*

That is, $\widetilde{M}$ consists of **two copies** of $\overline{M}$ glued along the boundary.

# The gluing maps

Near $\partial M$, one constructs $W^{2,\infty}$ charts called **pseudo-geodesic coordinates**: for each $X_0 \in \partial M$ there is a diffeomorphism $\psi'_j : U'_j \to \mathbb{R}^d$ with

$$\Delta = \frac{1}{\kappa(y,z)} \nabla^{\top}_{y,z} \kappa(y,z)\, b(y,z)\, \nabla_{y,z}, \qquad b\big|_{\partial\Omega} = \begin{pmatrix} 1 & 0 \\ 0 & b'(z) \end{pmatrix}.$$

The metric is *diagonal at the boundary*: tangent to a geodesic coordinate system.

Define the **gluing map**:

$$\Psi_j(x,\varepsilon) = \begin{cases} \psi'_j(x), & \varepsilon = +1, \\ S \circ \psi'_j(x), & \varepsilon = -1, \end{cases} \qquad S(s,z) = (-s,z).$$

$S$ is the reflection operator across $\{s = 0\}$.

# Theorem: the double manifold

## Theorem (Double manifold, [BM23)

*] Let $g$ be a $W^{1,\infty}$ metric on a $W^{2,\infty}$ manifold $M$ with boundary. There exist a $W^{2,\infty}$ structure on $\widetilde{M}$, a metric $\widetilde{g} \in W^{1,\infty}$ and a density $\widetilde{\kappa} \in W^{1,\infty}$ on $\widetilde{M}$ such that:*

1. *The maps $\iota^{\pm} : M \ni x \mapsto (x, \pm 1) \in \widetilde{M}$ are isometric embeddings.*
2. *The density induced on each copy of $M$ equals $\kappa$.*
3. *For each eigenfunction $e$ with eigenvalue $\lambda^2$ of $(-\Delta, Dir/Neu)$ on $M$, there exists $\widetilde{e}$ with the same eigenvalue of $(-\widetilde{\Delta})$ on $\widetilde{M}$ such that*

$$\widetilde{e}\big|_{M\times\{+1\}} = e, \qquad \widetilde{e}\big|_{M\times\{-1\}} = \begin{cases} -e & \text{(Dirichlet)}, \\ +e & \text{(Neumann)}. \end{cases}$$

# Eigenfunction extension

**Dirichlet case** ($e|_{\partial M} = 0$): setting $\widetilde{e}(x, \pm 1) = \pm e(x)$ is well-defined since $e = 0 = -e$ on $\partial M$. In local coordinates near $x \in \partial M$:

$$\widetilde{e}(s, z) = \begin{cases} e(s, z), & s > 0, \\ -e(-s, z), & s < 0. \end{cases}$$

Odd reflection: $\widetilde{e}$ is continuous and satisfies the equation.

**Neumann case** ($\partial_\nu e|_{\partial M} = 0$): set $\widetilde{e}(x, \pm 1) = +e(x)$. In coordinates:

$$\widetilde{e}(s, z) = \begin{cases} e(s, z), & s > 0, \\ e(-s, z), & s < 0. \end{cases}$$

Even reflection: the normal derivative cancels by symmetry.

## Remark

*In both cases, $-\widetilde{\Delta}\,\widetilde{e} = \lambda^2 \widetilde{e}$ on $\widetilde{M}$. The eigenvalue is preserved.*

# Consequence: spectral inequalities with boundary

Combining the theorem with the boundaryless results:

## Corollary

*The spectral inequality for boundaryless manifolds,*

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \;\Rightarrow\; \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \sup_{x \in E_2} |\phi(x)|,$$

*implies the same inequality for Dirichlet or Neumann boundary conditions on $M$.*

## Remark

*Since the Dirichlet/Neumann orthonormal bases extend as above, the generated space is dense in $L^2(\widetilde{M})$, so there exists a Hilbert basis of $L^2(\widetilde{M})$ consisting of eigenfunctions of $\widetilde{\Delta}$.*

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# Problem statement

Let $\Omega \subset \mathbb{R}^d$ be a smooth bounded domain, $\omega \subset \Omega$ with $|\omega| > 0$, $T > 0$.

**Simultaneous null controllability problem**: find $f \in L^2((0,T) \times \omega)$ such that the solutions of

$$\begin{cases} (\partial_t - \Delta)u = f\,\mathbf{1}_{(0,T)\times\omega}, & u\big|_{\partial\Omega} = 0, \quad u\big|_{t=0} = u_0, \\ (\partial_t - \Delta)v = f\,\mathbf{1}_{(0,T)\times\omega}, & \partial_\nu v\big|_{\partial\Omega} = 0, \quad v\big|_{t=0} = v_0, \end{cases}$$

satisfy $u|_{t>T} = 0$ and $v|_{t>T} = 0$.

## Remark

*The interest lies in the fact that both equations have independent dynamics and yet they can be steered to zero using exactly the same control $f$.*

# Main results

**Theorem ([BM23)**

*] Let $T > 0$ and $\omega \subset \Omega$ with $|\omega| > 0$. Then the heat equation in $\Omega$ is* **simultaneously null controllable** *with Dirichlet and Neumann boundary conditions.*

The proof relies on the following simultaneous spectral inequality:

**Theorem ([BM23)**

*] Let $\omega \subset \Omega$ with $|\omega| \geq m$. There exist $C, D > 0$ such that for all $\Lambda > 0$ and $u, v \in L^2(\Omega)$:*

$$\|\Pi_\Lambda^D u\|_{L^\infty(\Omega)} + \|\Pi_\Lambda^N v\|_{L^\infty(\Omega)} \leq Ce^{C\Lambda} \|\mathbf{1}_\omega \left(\Pi_\Lambda^D u + \Pi_\Lambda^N v\right)\|_{L^1(\omega)}.$$

# Proof of the simultaneous spectral inequality

Let $u, v \in L^2(M)$. Define $\widetilde{u} \in L^2(\widetilde{M})$ by

$$\widetilde{u}(x,+1) = (u+v)(x), \qquad \widetilde{u}(x,-1) = (-u+v)(x).$$

By the eigenfunction extension theorem:

$$\widetilde{\Pi}_\Lambda(\widetilde{u})(\cdot,+1) = \Pi^D_\Lambda u + \Pi^N_\Lambda v, \qquad \widetilde{\Pi}_\Lambda(\widetilde{u})(\cdot,-1) = -\Pi^D_\Lambda u + \Pi^N_\Lambda v.$$

Apply the spectral inequality on $\widetilde{M}$ (boundaryless) with $\widetilde{\omega} = \omega \times \{+1\}$:

$$\begin{aligned} \|\Pi^D_\Lambda u\|^2_{L^\infty} + \|\Pi^N_\Lambda v\|^2_{L^\infty} &= \|\widetilde{\Pi}_\Lambda \widetilde{u}\|^2_{L^\infty(\widetilde{M})} \\ &\leq Ce^{C\Lambda} \|\mathbf{1}_{\widetilde{\omega}}\, \widetilde{\Pi}_\Lambda \widetilde{u}\|^2_{L^1(\widetilde{\omega})} \\ &= Ce^{C\Lambda} \|\mathbf{1}_\omega (\Pi^D_\Lambda u + \Pi^N_\Lambda v)\|^2_{L^1(\omega)}. \quad \square \end{aligned}$$

# Proof of controllability pt. 1

Given $u, v \in L^2(M)$, define $\widetilde{u}(x, \pm 1)$ as above and set $\widetilde{\omega} = \omega \times \{+1\}$.

By the Burq–Moyano controllability result on $\widetilde{M}$ (boundaryless), there exists $\widetilde{f} \in L^2((0,T) \times \widetilde{\omega})$ such that the solution $\widetilde{U}$ of

$$(\partial_t - \widetilde{\Delta})\widetilde{U} = \widetilde{f}\,\mathbf{1}_{(0,T)\times\widetilde{\omega}}, \quad \widetilde{U}\,|_{t=0} = \tilde{u}, \quad \widetilde{U}\big|_{t=0} = \widetilde{u}_0,$$

satisfies $\widetilde{U}|_{t\geq T} = 0$.

**Descent to** $M$: define

$$u(t,x) = \widetilde{U}(t,x,+1) - \widetilde{U}(t,x,-1), \quad v(t,x) = \widetilde{U}(t,x,+1) + \widetilde{U}(t,x,-1).$$

By construction, $u$ satisfies Dirichlet and $v$ satisfies Neumann boundary conditions.

# Proof of controllability pt. 2

**Neumann condition**: in normal coordinates near $\partial M$ with $M = \{x_n > 0\}$, the gluing gives

$$\partial_\nu v\big|_{x_n=0} = \partial_{x_n}\Big(\widetilde{U}(\cdot,+1) + \widetilde{U}(\cdot,-1)\Big)\big|_{x_n=0} = \partial_{x_n}\widetilde{U}(\cdot,+1) - \partial_{x_n}\widetilde{U}(\cdot,-1) = 0.$$

**Equation satisfied**: since $\widetilde{f}(\cdot,-1)\,\mathbf{1}_{\widetilde{\omega}} = 0$ by the choice $\widetilde{\omega} = \omega \times \{+1\}$,

$$(\partial_t - \Delta)u = \widetilde{f}(t,x,+1)\,\mathbf{1}_{(0,T)\times\omega} - \widetilde{f}(t,x,-1)\mathbf{1}_{(0,T)\times\widetilde{\omega}} = f(t,x)\,\mathbf{1}_{(0,T)\times\omega},$$

$$(\partial_t - \Delta)v = \widetilde{f}(t,x,+1)\,\mathbf{1}_{(0,T)\times\omega} + \widetilde{f}(t,x,-1)\mathbf{1}_{(0,T)\times\widetilde{\omega}} = f(t,x)\,\mathbf{1}_{(0,T)\times\omega}.$$

Thus $u$ and $v$ solve the system with the *same* control $f$.

Finally, $\widetilde{U}|_{t\geq T} = 0$ implies $u|_{t\geq T} = 0$ and $v|_{t\geq T} = 0$. □

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# Summary

1. **Double manifold** $\widetilde{M}$: two copies of $M$ glued via $W^{2,\infty}$ pseudo-geodesic charts. The metric $\widetilde{g}$ is $W^{1,\infty}$.
2. **Eigenfunction extension**: odd reflection (Dirichlet) or even reflection (Neumann) preserves eigenvalues. The basis of $L^2(M)$ extends to a basis of $L^2(\widetilde{M})$.
3. **Spectral inequalities with boundary**: deduced from the boundaryless results on $\widetilde{M}$ via the isometric embeddings $\iota^{\pm}$.
4. **Simultaneous controllability**: reduced to a single problem on $\widetilde{M}$. Projecting back to $M$ yields two systems controlled by the same $f$.


N. Burq and C. Moyano, *Propagation of smallness and control for heat equations*, 2023.